

The implicit function theorem and the inverse mapping theorem

1. Given the function $F(x, y, z) := xy + y^2 + z^2 - e^z - 4$.

- (a) Show that the equation $F(x, y, z) = 0$ defines around the point $(0, e, 2)$ a function $z = f(x, y)$.

The function F is of class $\mathcal{C}^\infty$ and $F(0, e, 2) = 0$ while

$$F_z(x, y, z) = 2z - e^z \quad \Rightarrow \quad F_z(0, e, 2) = 4 - e^2 \neq 0.$$

Hence, by the IFT there exist $\delta > 0$, $\beta > 0$ and a unique function

$$f : B((0, e), \delta) \rightarrow (2 - \beta, 2 + \beta) \quad \text{of class } \mathcal{C}^\infty \text{ such that}$$

$$\forall (x, y, z) \in B((0, e), \delta) \times (2 - \beta, 2 + \beta), \quad F(x, y, z) = 0 \quad \Leftrightarrow \quad z = f(x, y).$$

- (b) Find $\nabla f(0, e)$.

By the IFT,

$$f_x(x, y) = -\frac{F_x(x, y, f(x, y))}{F_z(x, y, f(x, y))} \quad \text{and} \quad f_y(x, y) = -\frac{F_y(x, y, f(x, y))}{F_z(x, y, f(x, y))} \quad \forall (x, y) \in B((0, e), \delta).$$

Now,

$$F_x = y, \quad F_y = x + 2y \quad \Rightarrow \quad \begin{cases} f_x(0, e) = -\frac{e}{4 - e^2}, \\ f_y(0, e) = -\frac{2e}{4 - e^2}. \end{cases}$$

So,

$$\nabla f(0, e) = \left(\frac{e}{4 - e^2}, -\frac{2e}{4 - e^2} \right).$$

2. Given the function $F(x, y) := 2 \sin x - e^{2x} \ln(e + 2xy) + e^y \ln(e - 2xy)$.

- (a) Around which of the points $(0, \pi/2)$ and $(0, 0)$ does the equation $F(x, y) = 0$ define a unique function $y = f(x)$? **Apply IFT!**

- (b) Find the limit $\lim_{x \rightarrow 0} \frac{f(x)}{x^2}$. **Apply L'H's rule!**

3. Let $F \in \mathcal{C}^1(\mathbb{R}^2)$ be such that

$$F(x, y) = 0 \quad \Leftrightarrow \quad y^2 = (\arctan x)^2. \quad (*)$$

Use the implicit function theorem to show that $\nabla F(0, 0) = (0, 0)$.

By contradiction, if $\nabla f(0, e) \neq (0, 0)$, for instance $F_y(0, 0) \neq 0$ while we know that $F(0, 0) = 0$ since $0^2 = (\arctan(0))^2$. Then, by the IFT there exist $\delta > 0$, $\beta > 0$ and a **unique function**

$f : B(-\delta, \delta) \rightarrow (-\beta, \beta)$ of class $\mathcal{C}^1$ such that

$$\forall (x, y) \in (-\delta, \delta) \times (-\beta, \beta), \quad F(x, y) = 0 \quad \Leftrightarrow \quad y = f(x)$$

which contradicts (*). **Check the contradiction!**

4. Let Γ be the set of points $(x, y, s, t) \in \mathbb{R}^4$ which satisfy the system of equations:

$$\begin{cases} F(x, y, s, t) := \sin(x + y) + 1 - e^{s+t} + s^3 t^2 = 0 \\ G(x, y, s, t) := \cos(st) - 1 + \arctan(2x + y) + t = 0. \end{cases}$$

(i) Show that, in a neighborhood of $(0, 0, 0, 0)$, the set Γ defines implicitly two functions $s = f(x, y)$ and $t = g(x, y)$.

Notice that the function F and G are of class $\mathcal{C}^\infty$. Also

$$\begin{cases} F(0, 0, 0, 0) = 0 \\ G(0, 0, 0, 0) = 0 \end{cases}$$

Now, we need to check whether

$$\det \left(\frac{\partial(F, G)}{\partial(s, t)}(0, 0, 0, 0) \right) = \begin{vmatrix} F_s(0, 0, 0, 0) & F_t(0, 0, 0, 0) \\ G_s(0, 0, 0, 0) & G_t(0, 0, 0, 0) \end{vmatrix} \neq 0.$$

$$F_s = -e^{s+t} + 3s^2 t^2, \quad F_t = -e^{s+t} + 2s^3 t, \quad G_s = -t \sin(st), \quad G_t = -s \sin(st) + 1.$$

$$\begin{cases} F(0, 0, 0, 0) = 0 \\ G(0, 0, 0, 0) = 0 \end{cases}$$

Now, we need to check whether

$$\det \left(\frac{\partial(F, G)}{\partial(s, t)}(0, 0, 0, 0) \right) = \begin{vmatrix} -1 & -1 \\ 0 & 1 \end{vmatrix} = -1 \neq 0.$$

Hence, there exist $\delta > 0$ and $\beta > 0$ and a unique function

$(f, g) : B((0, 0), \delta) \rightarrow B((0, 0), \beta)$ of class $\mathcal{C}^1$ such that

$$\forall (x, y) \in B((0, 0), \delta), \quad \forall (s, t) \in B((0, 0), \beta), \quad \begin{cases} F(x, y, s, t) = 0 \\ G(x, y, s, t) = 0. \end{cases} \Rightarrow \begin{cases} s = f(x, y), \\ t = g(x, y). \end{cases}$$

- (ii) Find the cartesian equations of the tangent planes to the graphs of the functions $s = f(x, y)$ and $t = g(x, y)$ at the point $(0, 0, 0)$.

We have the equations

$$s = f(0, 0) + f_x(0, 0)(x-0) + f_y(0, 0)(y-0) \text{ and } t = g(0, 0) + g_x(0, 0)(x-0) + g_y(0, 0)(y-0).$$

We will need

$$F_x = \cos(x+y), \quad F_y = \cos(x+y), \quad G_x = \frac{2}{1+(2x+y)^2}, \quad G_y = \frac{1}{1+(2x+y)^2}.$$

Using theorem 1.1 in the lecture notes L17, we find that

$$\begin{aligned} \frac{\partial(f, g)}{\partial(x, y)}(0, 0) &= - \left[\frac{\partial(F, G)}{\partial(s, t)}(0, 0, 0, 0) \right]^{-1} \left[\frac{\partial(F, G)}{\partial(x, y)}(0, 0, 0, 0) \right] \\ \Leftrightarrow \begin{pmatrix} f_x(0, 0) & f_y(0, 0) \\ g_x(0, 0) & g_y(0, 0) \end{pmatrix} &= - \begin{pmatrix} -1 & -1 \\ 0 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 1 & 1 \\ 2 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 2 & 1 \end{pmatrix} \\ \Leftrightarrow \begin{pmatrix} f_x(0, 0) & f_y(0, 0) \\ g_x(0, 0) & g_y(0, 0) \end{pmatrix} &= \begin{pmatrix} 3 & 2 \\ -2 & -1 \end{pmatrix} \end{aligned}$$

Noticing that $f(0, 0) = 0$ and $g(0, 0) = 0$, we find that

$$\boxed{s = 3x + 2y \quad \text{and} \quad t = -2x - y}.$$

5. Consider the functions

$$F(x, y, z) := -x - 3y + \arctan(z-1) + \sin z + \frac{\pi}{4} \quad \text{and} \quad G(x, y, z) := x + \cos y - e^z \quad \forall (x, y, z) \in \mathbb{R}^3.$$

- (a) Show that the system of equations

$$\begin{cases} F(x, y, z) = 0 \\ G(x, y, z) = 0 \end{cases}$$

defines around the point $(0, 0, 0)$ a unique regular curve $\mathcal{C}$ with parametric equations

$$\begin{cases} y = f(x), \\ z = g(x). \end{cases}$$

Notice that the function F and G are of class $\mathcal{C}^\infty$. Also

$$\begin{cases} F(0, 0, 0) = 0 \\ G(0, 0, 0) = 0 \end{cases} \quad \text{and} \quad \frac{\partial(F, G)}{\partial(y, z)}(x, y, z) = \begin{vmatrix} F_y(x, y, z) & F_z(x, y, z) \\ G_y(x, y, z) & G_z(x, y, z) \end{vmatrix}.$$

$$F_y = -1, \quad F_z = \frac{1}{1+(z-1)^2} + \cos z, \quad G_y = -\sin y, \quad G_z = -e^z.$$

Hence,

$$\frac{\partial(F, G)}{\partial(y, z)}(0, 0, 0) = \begin{vmatrix} F_y(0, 0, 0) & F_z(0, 0, 0) \\ G_y(0, 0, 0) & G_z(0, 0, 0) \end{vmatrix} = \begin{vmatrix} -1 & 3/2 \\ 0 & -1 \end{vmatrix} = 1 \neq 0.$$

Hence, by the implicit function theorem for system of equations (Theorem 1.1 in the lecture notes L17), there exist $\delta > 0$ and $\beta > 0$ and a unique vector function

$$(f, g) : (-\delta, \delta) \rightarrow B((0, 0), \beta)$$

of class $\mathcal{C}^\infty$ such that

$$\forall x \in (-\delta, \delta), \forall (y, z) \in B((0, 0), \beta), \quad \begin{cases} F(x, y, z) = 0 \\ G(x, y, z) = 0 \end{cases} \Leftrightarrow \begin{cases} y = f(x) \\ z = g(x) \end{cases}$$

Therefore, we obtain the curve $\mathcal{C}$ with parametric equations

$$y = f(x), \quad z = g(x) \quad \text{with} \quad x \in (-\delta, \delta)$$

or described by the vector function $\vec{r} : (-\delta, \delta) \rightarrow \mathbb{R}^3$

$$\vec{r}(x) := (x, f(x), g(x)) \quad \forall x \in (-\delta, \delta).$$

- (b) Write down the parametric equations of the tangent line to the curve $\mathcal{C}$ at the point $(0, 0)$.

From 2AC, the tangent line to the curve $\mathcal{C}$ at the point $(0, 0, 0)$ has direction $\vec{r}'(0)$. We find that

$$\vec{r}'(x) = (1, f'(x), g'(x)).$$

We know from Theorem 1.1 in the lecture notes L17 that

$$\begin{pmatrix} f'(x) \\ g'(x) \end{pmatrix} = - \left[\frac{\partial(F, G)}{\partial(y, z)}(x, f(x), g(x)) \right]^{-1} \begin{pmatrix} F_x(x, f(x), g(x)) \\ G_x(x, f(x), g(x)) \end{pmatrix}$$

We find that

$$\begin{aligned} F_x = -1, \quad G_x = 1 &\Rightarrow \begin{pmatrix} f'(0) \\ g'(0) \end{pmatrix} = \begin{pmatrix} -1 & 3/2 \\ 0 & -1 \end{pmatrix}^{-1} \begin{pmatrix} -1 \\ 1 \end{pmatrix} \\ &= \frac{1}{1} \begin{pmatrix} -1 & -3/2 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} -1 \\ 1 \end{pmatrix} = \begin{pmatrix} -1/2 \\ -1 \end{pmatrix}. \end{aligned}$$

So, the tangent line to the curve $\mathcal{C}$ at the point $(0, 0, 0)$ has direction

$$\vec{r}'(0) = (1, -1/2, -1).$$

Therefore, the tangent line to the curve $\mathcal{C}$ at the point $(0, 0, 0)$ has parametric equations

$$x = \lambda, \quad y = -\frac{1}{2}\lambda, \quad z = -\lambda, \quad \lambda \in \mathbb{R}.$$

6. Let $f = (f_1, f_2) : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a function such that

$$\begin{cases} f_1(x, y) := \sin(\frac{\pi}{2}x - y) + e^{x-2y} + y^2 - x, \\ f_2(x, y) := \arctan(e^{x-2y}) - y - \frac{x}{2} \end{cases} \quad \forall (x, y) \in \mathbb{R}^2.$$

- (a) Show that f is locally invertible around the point $(0, 0)$.
(b) Find the jacobian matrix of f^{-1} at the point $(1, \frac{\pi}{4})$.

Solve as in the notes!